

cells - Answer Key

1. Describe the main differences between prokaryotic and eukaryotic cells.

Answer: Prokaryotic cells lack a membrane-bound nucleus and most organelles, whereas eukaryotic cells contain a true nucleus and a variety of specialized organelles. Prokaryotic cells are typically found in bacteria and archaea, while eukaryotic cells are found in plants, animals, fungi, and protists. The presence of a nucleus and organelles in eukaryotic cells allows for greater complexity and specialization of cellular functions.

Explanation: The question requires a description of the main differences between prokaryotic and eukaryotic cells, including their structure and composition.

2. Describe the process of mitosis, including its main phases.

Answer: Mitosis is a process of cell division that produces two genetically identical daughter cells. It consists of four main phases: prophase, metaphase, anaphase, and telophase. During prophase, the chromosomes condense and the nuclear envelope breaks down. In metaphase, the chromosomes align at the center of the cell. In anaphase, the sister chromatids separate and move to opposite poles of the cell. In telophase, the nuclear envelope reforms and the chromosomes uncoil. This process is preceded by interphase, during which the cell grows and replicates its DNA.

Explanation: The text describes the process of mitosis and its main phases.

3. Describe the three main stages of cellular respiration and their functions.

Answer: The three main stages of cellular respiration are glycolysis, the Krebs cycle, and the electron transport chain. Glycolysis is the first stage, which occurs in the cytoplasm and produces a small amount of ATP. The Krebs cycle occurs in the mitochondrial matrix and produces more ATP. The electron transport chain is the final stage, which produces a large majority of the cell's ATP through chemiosmosis. These stages work together to convert nutrients into usable energy for the cell.

Explanation: This question requires the test-taker to describe the three main stages of cellular respiration and their functions, demonstrating their understanding of the process.